

Programme Status

Summary Retrieval quality degrades non-linearly as the corpus grows past the point where the embedding model was calibrated, and the failure is quiet: recall stays flat while precision collapses in the tail. Teams that measure only average relevance will not see it. The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.

Risks 🕒 Index rebuild window is contended.
🕒 Reranker training data is stale by two quarters. 🕒 Cold-start latency is unmeasured in production.

Risk	Impact	Owner
Rebuild	High	A
Staleness	Medium	B
Cold start	Medium	C

The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.